

SHIONOMISAKI, Japan

WMO: 477780

Lat: 33.4503N

Lon: 135.7572E

Elev: 69

StdP: 100.50

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	2.9	-8.2	1.9	4.6	-7.0	2.1	5.0	12.3	11.5	10.9	10.9	5.0	310	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	30.6	26.1	29.9	25.9	29.2	25.6	27.0	29.3	26.6	28.7	26.2	28.2	4.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.4	22.1	28.4	25.9	21.5	28.0	25.5	20.9	27.7	85.6	29.3	83.6	28.9	81.8	28.4	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.4	8.4	0.0	32.1	1.3	0.7	-0.9	32.6	-1.7	33.0	-2.4	33.4	-3.4	33.9		
			-2.8	27.6	1.1	0.5	-3.6	28.0	-4.2	28.4	-4.8	28.7	-5.6	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	8.1	8.8	11.6	15.7	19.5	22.2	26.0	27.2	24.8	20.5	15.5	10.6
	DBStd	6.95	2.66	3.04	3.03	2.66	1.82	1.89	1.71	1.34	2.00	2.31	2.88	3.00
	HDD10.0	173	71	54	18	1	0	0	0	0	0	0	1	29
	HDD18.3	1225	317	267	210	84	9	0	0	0	0	7	91	240
	CDD10.0	2946	12	21	67	172	296	367	497	534	443	324	167	48
	CDD18.3	956	0	0	0	6	47	117	238	276	193	73	7	0
	CDH23.3	6558	0	0	0	1	42	343	1986	2753	1272	158	4	0
	CDH26.7	1289	0	0	0	0	0	13	364	715	193	4	0	0
Wind	WSAvg	4.2	4.4	4.4	4.5	4.3	4.2	4.2	4.1	4.1	4.2	4.1	3.9	4.2
Precipitation	PrecAvg	2647	98	111	174	232	258	373	277	273	314	250	178	98
	PrecMax	3622	291	280	378	475	667	595	743	1049	855	956	537	309
	PrecMin	1736	0	19	47	85	99	91	29	35	65	30	21	0
	PrecStd	463	63	65	72	103	97	119	177	171	163	154	126	73
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	18.8	20.1	23.1	25.6	28.1	31.3	31.7	30.4	27.2	23.7	20.1
		MCWB	15.1	16.5	15.4	17.9	20.3	24.5	25.8	26.7	25.5	23.0	20.1	16.9
	2%	DB	15.7	17.4	19.0	21.8	24.4	26.8	30.2	30.9	29.4	26.1	22.2	18.0
		MCWB	12.4	14.8	15.6	17.5	19.9	23.7	26.0	26.4	25.0	22.2	18.4	14.2
	5%	DB	13.8	15.7	18.0	20.8	23.6	25.9	29.4	30.2	28.6	25.1	21.0	16.5
		MCWB	9.7	12.3	14.6	17.1	19.4	23.2	25.8	26.1	24.6	21.3	17.3	12.5
	10%	DB	12.3	13.8	16.8	19.8	22.8	25.2	28.6	29.6	27.8	24.0	20.0	15.2
		MCWB	8.2	9.9	13.2	16.1	18.8	22.7	25.5	25.9	24.2	20.2	16.3	11.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	17.2	18.1	20.7	23.3	25.6	27.2	28.0	26.9	24.9	21.6	18.1
		MCDB	17.2	18.3	18.9	21.4	24.3	26.8	29.5	30.1	28.9	26.2	22.8	19.8
	2%	WB	13.2	15.4	16.7	19.4	22.0	24.9	26.7	27.3	26.1	23.5	19.6	15.2
		MCDB	15.0	17.1	18.1	20.5	23.0	26.0	28.9	29.5	28.1	25.2	21.2	17.2
	5%	WB	10.6	13.1	15.2	18.1	20.8	24.3	26.2	26.8	25.6	22.3	18.2	13.4
		MCDB	12.7	15.0	17.4	19.8	22.4	25.5	28.2	29.1	27.6	24.3	20.2	15.7
	10%	WB	8.9	10.8	13.7	16.9	19.8	23.5	25.9	26.5	25.1	21.1	17.0	11.9
		MCDB	11.7	13.0	16.3	19.1	21.7	24.8	27.7	28.7	27.0	23.3	19.2	14.6
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.4	6.6	6.1	5.4	4.4	4.1	4.4	4.7	5.2	5.7	5.8
		MCDBR	7.4	7.3	6.9	6.5	6.1	4.8	4.8	4.9	5.1	5.6	6.2	6.8
		MCWBR	6.5	7.2	6.5	5.1	4.4	3.1	2.6	2.7	3.1	4.3	5.3	6.0
	5% WB	MCDBR	6.8	7.2	6.6	5.4	4.8	3.9	4.2	4.5	4.5	4.9	5.8	6.3
MCWBR		7.3	7.8	7.3	5.3	4.4	2.9	2.5	2.5	3.1	4.6	5.9	6.6	
Clear-Sky Solar Irradiance	taub	0.393	0.440	0.500	0.532	0.540	0.547	0.545	0.513	0.487	0.465	0.420	0.387	
	taud	2.210	2.091	1.941	1.911	1.967	2.037	2.081	2.173	2.210	2.180	2.218	2.244	
	Ebn at Noon	796	794	776	772	769	761	760	779	781	762	761	779	
	Edh at Noon	115	143	179	193	184	171	163	147	136	129	113	105	
All-Sky Solar Radiation	RadAvg	2.64	3.27	4.26	4.91	5.20	4.46	5.38	5.53	4.31	3.33	2.71	2.40	
	RadStd	0.21	0.26	0.30	0.55	0.58	0.47	0.61	0.43	0.35	0.35	0.28	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page